2 **(a)** The Williamson ether synthesis involves nucleophilic substitution between a halogenoalkane (RX) and alkoxides (RO^-), the conjugate base of an alcohol. The alkoxide serves as the nucleophile in the reaction. An example of the reaction can be seen below:

$$RO^- + R'X \longrightarrow ROR' + X^-$$

A solution containing CH_3ONa, is reacted separately with 1-bromopropane and 2-bromo-2-methylpropane.

(i) Predict the predominant mechanism for:

I) the reaction of 1-bromopropane with CH_3ONa

II) the reaction of 2-bromo-2-methylpropane with CH_3ONa

Explain your reasoning. [3]

(ii) For each mechanism, state and explain the stereochemical outcome of the nucleophilic substitution reaction. [2]

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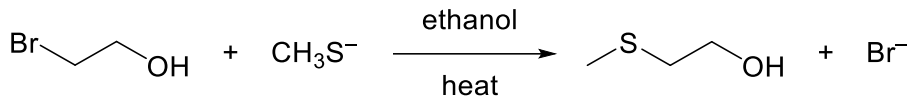
Question 2 continues on the next page.

(b) Thiolates such as methyl thiolate, CH_3S^-, act similarly to alkoxides in the nucleophilic substitution of halogenoalkanes, forming a sulfide.

The synthesis of 2-hydroxyethyl methyl sulfide from 2-bromoethanol using CH_3S^- can be seen from the following reaction scheme:



(i) Describe a simple chemical test to distinguish between 2-bromoethanol and 2-hydroxyethyl methyl sulfide. [2]

The kinetics of the reaction was studied with the results given in Table 2.1.

Table 2.1

experiment	$[CH_3S^-]$ / mol dm^{-3}	$[CH_2BrCH_2OH]$ / mol dm^{-3}	relative rate
1	0.100	0.150	1.00
2	0.150	0.150	1.50
3	0.200	0.200	2.67

(ii) Define the term *order of reaction*. [1]

(iii) Use the data to determine the order of reaction with respect to both CH_3S^- and CH_2BrCH_2OH. [2]

(iv) Hence, write a rate equation for the reaction. [1]

(v) Using your answer in **(b)(iv)**, describe the mechanism for the reaction. [3]

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(c) Sodium methoxide, CH_3ONa, can also be used to synthesis 2-hydroxyethyl methyl sulfide from 2-bromoethanol. The rate of reaction was found to be slower than when sodium methanethiol, CH_3SNa, is used.

By considering the strength of the nucleophile, suggest why the rate of reaction is slower when using sodium methoxide. [1]

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(d) The rate of reaction for the synthesis increases when the temperature is increased. With the aid of a clearly labelled Maxwell-Boltzmann distribution curve, explain this observation. [3]

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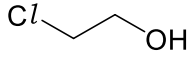
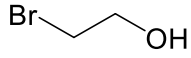
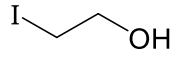
(e) Another series of experiments were done to study the difference in the rates of nucleophilic substitution for the following compounds using methyl thiolate.



2-chloroethanol



2-bromoethanol



2-iodoethanol

Deduce the order of increasing rate of reaction of the following compounds. Explain your answer. [2]

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[Total: 20]